

Chapter 7: QML Technique 1

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May 20, 2022

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- Quantum Fourier Transform (QFT)
- Quantum Phase Estimation (QPE)

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- Example

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Quantum Fourier Transform (QFT)

- The QFT is the quantum implementation of the discrete Fourier transform.
- The Hadamard gate $H := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ can be regarded as the 2-qubit QFT.

$$\sum_{k=0}^{N-1} r_k |k\rangle \xrightarrow{QFT} \sum_{k=0}^{N-1} y_k |k\rangle, \quad y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} r_j e^{2\pi i \frac{kj}{N}}.$$

$$\begin{aligned} \text{QFT}|x\rangle &= \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{2\pi i \frac{xk}{2^n}} |k\rangle \\ &= \frac{1}{2^{n/2}} \bigotimes_{k=1}^n \left(|0\rangle + e^{2\pi i \frac{x}{2^k}} |1\rangle \right) \\ &= \frac{1}{2^{n/2}} \left(|0\rangle + e^{2\pi i \frac{x}{2}} |1\rangle \right) \otimes \left(|0\rangle + e^{2\pi i \frac{x}{2^2}} |1\rangle \right) \otimes \cdots \otimes \left(|0\rangle + e^{2\pi i \frac{x}{2^n}} |1\rangle \right). \end{aligned}$$

$$\begin{aligned}
 & \text{QFT} |x_1 x_2 \dots x_{2^n-1}\rangle \\
 &= \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{2\pi i \frac{xk}{2^n}} |k\rangle \\
 &= \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} \prod_{j=1}^n e^{2\pi i \frac{xk_j}{2^j}} |k_1 k_2 \dots k_n\rangle \quad \left(k = \sum_{j=1}^n 2^{n-j} k_j \right) \\
 &= \frac{1}{2^{n/2}} \sum_{k_2=0}^1 \dots \sum_{k_n=0}^1 \prod_{j=2}^n e^{2\pi i \frac{x y_j}{2^j}} \left(|0\rangle + e^{2\pi i \frac{x}{2}} \right) \otimes |k_2\rangle \otimes \dots \otimes |k_n\rangle \\
 &= \frac{1}{2^{n/2}} \sum_{k_3=0}^1 \dots \sum_{k_n=0}^1 \prod_{j=3}^n e^{2\pi i \frac{x y_j}{2^j}} \left(|0\rangle + e^{2\pi i \frac{x}{2}} \right) \otimes \left(|0\rangle + e^{2\pi i \frac{x}{2^2}} \right) \otimes |k_3\rangle \otimes \dots \otimes |k_n\rangle \\
 &= \frac{1}{2^{n/2}} \bigotimes_{k=1}^n \left(|0\rangle + e^{2\pi i \frac{x}{2^k}} |1\rangle \right) \\
 &= \frac{1}{2^{n/2}} \left(|0\rangle + e^{2\pi i \frac{x}{2}} |1\rangle \right) \otimes \left(|0\rangle + e^{2\pi i \frac{x}{2^2}} |1\rangle \right) \otimes \dots \otimes \left(|0\rangle + e^{2\pi i \frac{x}{2^n}} |1\rangle \right).
 \end{aligned}$$

- The inverse QFT:

$$\text{QFT}^{-1}|x\rangle = \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{-2\pi i \frac{xk}{2^n}} |k\rangle.$$

$$\begin{aligned} \text{QFT}^{-1}\text{QFT}|x\rangle &= \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{2\pi i \frac{xk}{2^n}} \text{QFT}^{-1}|k\rangle \\ &= \frac{1}{2^n} \sum_{k=0}^{2^n-1} e^{2\pi i \frac{xk}{2^n}} \sum_{l=0}^{2^n-1} e^{-2\pi i \frac{kl}{2^n}} |l\rangle = \frac{1}{2^n} \sum_{k=0}^{2^n-1} \sum_{l=0}^{2^n-1} e^{2\pi i \frac{k}{2^n} (x-l)} |l\rangle \\ &= |x\rangle + \frac{1}{2^n} \sum_{\substack{l \neq x \\ 1 \leq l \leq 2^n-1}} \sum_{k=0}^{2^n-1} e^{2\pi i \frac{k}{2^n} (x-l)} |l\rangle = |x\rangle. \end{aligned}$$

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Quantum Phase Estimation (QPE)

- The QPE is a quantum algorithm to estimate the phase (or eigenvalue) of an eigenvector of a unitary operator.
- Let U be a unitary operator with an eigenvector $|\psi\rangle$ with its corresponding eigenvalue $e^{2\pi i\theta}$.

$$|\underbrace{00\cdots 0}_n\rangle \otimes |\psi\rangle \mapsto \frac{1}{2^n} \sum_{\ell=0}^{2^n-1} \sum_{k=0}^{2^n-1} e^{2\pi i k(\theta - \frac{\ell}{2^n})} |\ell\rangle \otimes |\psi\rangle.$$

- Five steps ¹: $|\psi_1\rangle \rightarrow |\psi_2\rangle \rightarrow |\psi_3\rangle \rightarrow |\psi_4\rangle \rightarrow |\psi_5\rangle$.
 - ① (Setup) $|\psi_1\rangle = |0\rangle^{\otimes n} \otimes |\psi\rangle$
 - ② (Superposition) Use the Hadamar gate $H^{\otimes n}$.

$$|\psi_2\rangle = \frac{1}{2^{n/2}}(|0\rangle + |1\rangle)^{\otimes n} \otimes |\psi\rangle.$$

¹[15] <https://learn.qiskit.org/course/ch-algorithms/quantum-phase-estimation>

Quantum Phase Estimation (QPE)

- ③ (Controlled Unitary Operations) Use the controlled unitary operator CU which activate as the unitary operator U on the target register only if its control qubit is $|1\rangle$.

$$\begin{aligned} |\psi_3\rangle &= \frac{1}{2^{\frac{n}{2}}} \left(|0\rangle + e^{2\pi i \theta 2^{n-1}} |1\rangle \right) \otimes \cdots \otimes \left(|0\rangle + e^{2\pi i \theta 2^0} |1\rangle \right) \otimes |\psi\rangle \\ &= \frac{1}{2^{\frac{n}{2}}} \sum_{k=0}^{2^n-1} e^{2\pi i \theta k} |k\rangle \otimes |\psi\rangle, \end{aligned}$$

where k denotes the integer representation of n -bit binary numbers.

- ④ (Inverse Fourier Transform) Apply the $\text{QFT}^\dagger$ to $|\psi_3\rangle$:

$$|\psi_4\rangle = \frac{1}{2^n} \sum_{\ell=0}^{2^n-1} \sum_{k=0}^{2^n-1} e^{2\pi i k \left(\theta - \frac{\ell}{2^n}\right)} |\ell\rangle \otimes |\psi\rangle.$$

- ⑤ (Measurement) The measurement in the computational basis gives the phase in the auxiliary register with high probability:

$$|\psi_5\rangle = |\text{observe the nearest integer to } 2^n \theta\rangle \otimes |\psi\rangle.$$

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HHL Algorithm description

- HHL algorithm is a method to solve linear systems introduced by three authors Harrow, Aram W., Avinatan Hassidim and Seth Lloyd² in 2009.
- (Problem setting) Consider the linear system

$$A|x\rangle = |b\rangle$$

- (Assumption) A is a Hermitian matrix whose spectral decomposition is

$$A = \sum_{j=0}^{N-1} \lambda_j |u_j\rangle\langle u_j|, \quad \lambda_j \in \mathbb{R}.$$

- (Assumption) A is invertible.

$$A^{-1} = \sum_{j=0}^{N-1} \frac{1}{\lambda_j} |u_j\rangle\langle u_j|, \quad \lambda_j \neq 0.$$

- (Solution) Combine above:

$$|x\rangle = A^{-1}|b\rangle = \sum_{j=0}^{N-1} \frac{b_j}{\lambda_j} |u_j\rangle.$$

²[8] Harrow, Aram W., Avinatan Hassidim and Seth Lloyd. *Quantum algorithm for linear systems of equations*. Physical review letters 103.15 (2009).

- (Additional assumption)³ Let κ be the condition number of A , or the ratio between A 's largest and smallest eigenvalues.

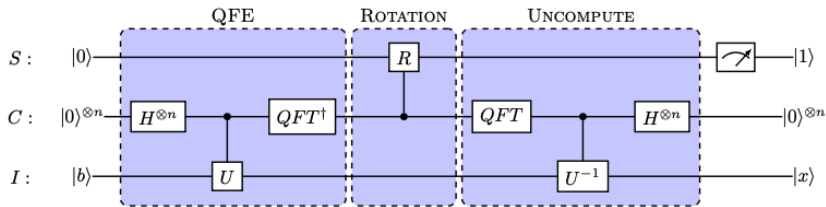
Assume that the singular values of A lie between 1 and $1/\kappa$; equivalently, $\kappa^{-2}I \leq A^\dagger A \leq I$.

In this case, this algorithm uses roughly $O(\kappa^2 \log(N)/\varepsilon)$ steps to output a state within distance (error) ε of $|x\rangle$.

³[8] Harrow, Aram W., Avinatan Hassidim and Seth Lloyd. *Quantum algorithm for linear systems of equations*. Physical review letters 103.15 (2009).

HHL Algorithm description

- The quantum circuit⁴:



Schematic diagram of HHL algorithm. The figure is adopted from [4].

- The top register S is the ancilla register, the middle one is the clock register and the bottom one is the input register.

⁴[4] Dervovic, D., Herbster, M., Mountney, P., Severini, S., Usher, N. and Wossnig, L., *Quantum linear systems algorithms: a primer*, (2018) arXiv preprint arXiv:1802.08227.

Five steps:⁵

- ① Loading the classical data $|b\rangle$
- ② We apply the QPE with the unitary matrix

$$U := e^{iAt} = \sum_{j=0}^{N-1} e^{i\lambda_j t} |u_j\rangle \langle u_j|.$$

Then, the quantum state of the register is

$$\begin{aligned} \text{QPE} \left(\sum_{j=0}^{2^n-1} b_j |0\rangle^{\otimes n_a} \otimes |u_j\rangle \right) &= \sum_{j=0}^{2^n-1} b_j \text{QPE} (|0\rangle^{\otimes n_a} \otimes |u_j\rangle) \\ &= \sum_{j=0}^{2^n-1} \underbrace{\left[\sum_{k=0}^{2^n-1} \sum_{l=0}^{2^n-1} \frac{1}{2^n} e^{2\pi i k \left(\frac{\lambda_j t}{2\pi} - \frac{l}{2^n} \right)} |l\rangle \right]}_{:= |\tilde{\lambda}_j\rangle} \otimes |u_j\rangle \end{aligned}$$

where $|\tilde{\lambda}_j\rangle$ is the n_a -bit binary representation of λ_j .

⁵[16] <https://learn.qiskit.org/course/ch-applications/solving-linear-systems-of-equations-using-hhl-and-its-qiskit-implementation>

- ③ Add an auxiliary qubit⁶ $|0\rangle$ and apply a rotation contained on $|\tilde{\lambda}_j\rangle$.

$$\sum_{j=0}^{2^n-1} b_j |\tilde{\lambda}_j\rangle \otimes |u_j\rangle \otimes \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j^2}} |0\rangle + \frac{C}{\tilde{\lambda}_j} |1\rangle \right)$$

where C is a constant for normalization which is smaller than the absolute value of the minimum of eigenvalues.

- ④ Uncompute to finish this algorithm.

$$\sum_{j=0}^{2^n-1} b_j |0\rangle \otimes |u_j\rangle \otimes \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j^2}} |0\rangle + \frac{C}{\tilde{\lambda}_j} |1\rangle \right)$$

- ⑤ Measure the ancilla qubit. If the measured quantity is $|1\rangle$, then the state qubit is

$$\sqrt{\frac{1}{\sum_{j=0}^{2^n-1} |b_j|^2 / |\tilde{\lambda}_j|^2}} \sum_{j=0}^{2^n-1} \frac{b_j}{\tilde{\lambda}_j} |0\rangle \otimes |u_j\rangle.$$

⁶ancilla qubit

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Example

- (Problem setting)⁷

$$Ax = b, \quad A = \frac{1}{4} \begin{pmatrix} 15 & 9 & 5 & -3 \\ 9 & 15 & 3 & -5 \\ 5 & 3 & 15 & -9 \\ 5 & 3 & 15 & -9 \\ -3 & -5 & -9 & 15 \end{pmatrix}, \quad b = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

- (Observation) A is Hermitian and has eigenvalues $\lambda_j = 2^{j-1}$ ($j = 1, 2, 3, 4$) with corresponding eigenvectors $|u_j\rangle$ given by

$$|u_1\rangle := -|00\rangle - |01\rangle - |10\rangle + |11\rangle = (-1 \quad -1 \quad -1 \quad 1)^T,$$

$$|u_2\rangle := +|00\rangle + |01\rangle - |10\rangle + |11\rangle = (1 \quad 1 \quad -1 \quad 1)^T,$$

$$|u_3\rangle := +|00\rangle - |01\rangle + |10\rangle + |11\rangle = (1 \quad -1 \quad 1 \quad 1)^T,$$

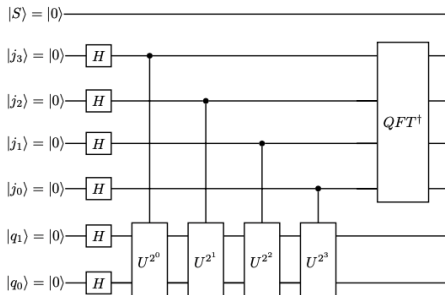
$$|u_4\rangle := -|00\rangle + |01\rangle + |10\rangle + |11\rangle = (-1 \quad 1 \quad 1 \quad 1)^T.$$

$$b = \sum_{j=1}^4 \beta_j |u_j\rangle, \quad \beta_j = \frac{1}{2}.$$

⁷[11] Dutta, S., Suau, A., Dutta, S., Roy, S., Behera, B. K. and Panigrahi, P. K., *Quantum circuit design methodology for multiple linear regression*. IET Quantum Communication, 1(2), (2020) 55-61. 

Example

- (Step 1, 2)



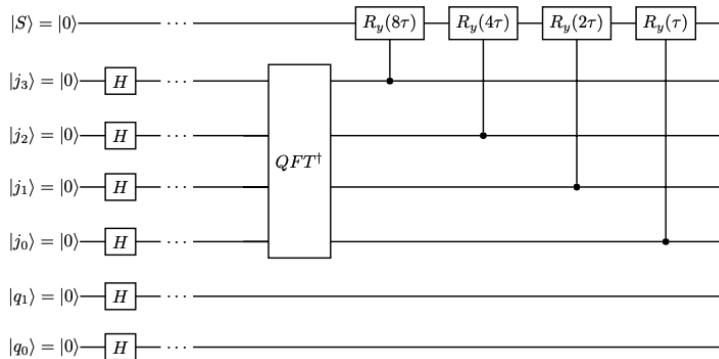
$$H \Rightarrow |b\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle.$$

$$\frac{1}{2}|0001\rangle^C \otimes |u_1\rangle^I + \frac{1}{2}|0010\rangle^C \otimes |u_2\rangle^I + \frac{1}{2}|0100\rangle^C \otimes |u_3\rangle^I + \frac{1}{2}|1000\rangle^C \otimes |u_4\rangle^I.$$

$\sum_{j=1}^4 \beta_j |u_j\rangle^I \otimes \left| \frac{\tilde{\lambda}_j t_0}{2\pi} \right\rangle^C$ under the assumption $t_0 = 2\pi$. Further, $|\tilde{\lambda}_j\rangle = |\lambda_j\rangle$

Example

- (Step 3)

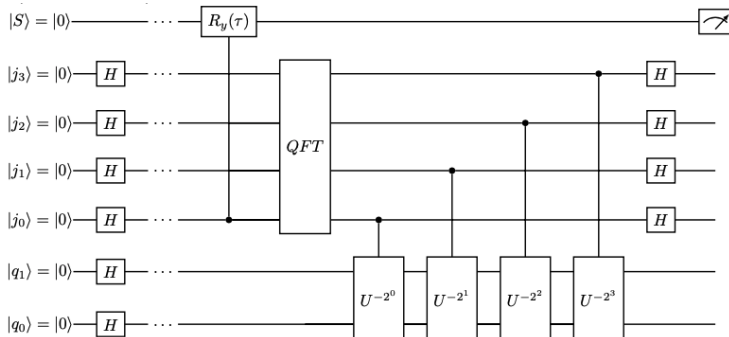


$\tau := \pi/2^r$. C is a number satisfying $|C| < \lambda_{min}$. Take $C = 8\pi/2^r$ using a a priori knowledge of the eigenvalues of A .

$$\sum_{j=1}^4 \beta_j |u_j\rangle^I \otimes |\tilde{\lambda}_j\rangle^C \otimes \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j^2}} |0\rangle + \frac{C}{\tilde{\lambda}_j} |1\rangle \right)^S$$

Example

- (Step 4, 5)



If $|1\rangle$ is measured in the ancilla qubit, then the state is

$$|00 \cdots 0\rangle \otimes C \sum_{j=1}^4 \frac{1/2}{2^{j-1}} |u_j\rangle$$

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Quadratic unconstrained binary optimization (QUBO)

- Quadratic unconstrained binary optimization (QUBO) is a combinatorial optimization problem with a wide range of applications from finance and economics to machine learning.⁸
- The QUBO problem is defined, for Q an $n \times n$ upper-triangular matrix of real weights and x a vector of n binary variables, as minimizing the function

$$f_Q(x) = \sum_{i=1}^n \sum_{j=1}^i Q_{ij} x_i x_j = \sum_{i=1}^n Q_{ii} x_i + \sum_{i < j} Q_{ij} x_i x_j$$

or, more concisely,

$$\min_{x \in \{0,1\}^n} x^T Q x.$$

⁸[19] https://en.wikipedia.org/wiki/Quadratic_unconstrained_binary_optimization ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

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- The Ising Model is a mathematical model used for studying phase transitions and other properties of physical systems that evolve in time.⁹
- The problem concerns a collection of n particles arranged on the vertices of a graph $G = (V, E)$, which is often assumed to be a d -dimensional grid.
- Each particle can be in one of two states, called spins, represented by ± 1 .

$$\mathcal{H}(\sigma) = \sum_{\langle i,j \rangle} J_{ij} \sigma_i \sigma_j - \mu \sum_j h_j \sigma_j$$

where real-valued parameters h_j, J_{ij}, μ for all i, j . J has the zero diagonal and symmetric. The spin variables σ_j are binary with values from $\{-1, +1\}$.

⁹[3] McGeoch, C. C., *Adiabatic quantum computation and quantum annealing: Theory and practice*, Synthesis Lectures on Quantum Computing, 5(2), (2014) 1-93.

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Relationship between Ising models and QUBO

- QUBOs are Ising models where the spin variables $\sigma_i \in \{-1, +1\}$ are transformed to binary-valued variables $x_i \in \{0, 1\}$.¹⁰ This transformation $\sigma \mapsto 2x - 1$ yields an equivalent QUBO problem:

$$\begin{aligned} f(x) &= \sum_{\langle i,j \rangle} -J_{ij}(2x_i - 1)(2x_j - 1) + \sum_j \mu h_j(2x_j - 1) \\ &= \sum_{\langle i,j \rangle} (-4J_{ij}x_i x_j + 2J_{ij}x_j + 2J_{ij}x_i - J_{ij}) + \sum_j (2\mu h_j x_j - \mu h_j) \quad \text{using } x_j = x_j x_j \\ &= \sum_{i=1}^n \sum_{j=1}^i Q_{ij} x_i x_j + C \end{aligned}$$

where

$$Q_{ij} = \begin{cases} -4J_{ij} & i \neq j \\ \sum_{\langle k,i \rangle} 2J_{ki} + \sum_{\langle i,l \rangle} 2J_{il} + 2\mu h_i & i = j \end{cases}, \quad C = -\sum_{\langle i,j \rangle} J_{ij} - \sum_j \mu h_j.$$

¹⁰Wiki-QUBO https://en.wikipedia.org/wiki/Quadratic_unconstrained_binary_optimization

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- [10]¹¹ “In 2016, access to the first cloud-based quantum computer became available, but noise and qubit limitations prevented serious implementations of the aforementioned quantum algorithms.

However, excitement grew about what could be done with these new devices, which have been called noisy intermediate-scale quantum (NISQ) computers.¹²

The key technological question is therefore how to make best use of today’s NISQ devices to achieve quantum advantage.

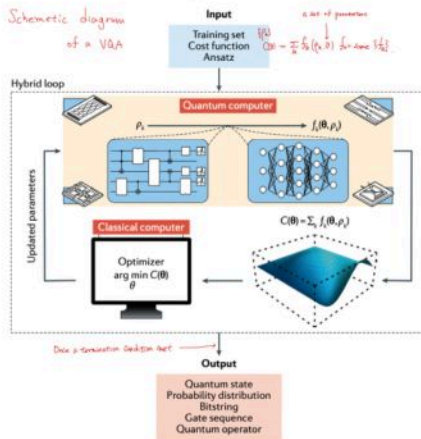
Variational quantum algorithms (VQAs) have emerged as the leading strategy to obtain quantum advantage on NISQ devices.”

¹¹[10] Cerezo, M., Arrasmith, A., Babbush, R., Benjamin, S. C., Endo, S., Fujii, K., ..., and Coles, P. J., *Variational quantum algorithms*. Nature Reviews Physics, 3(9), (2021) 625-644.

¹²[6] Preskill, J., *Quantum computing in the NISQ era and beyond*, Quantum 2, 79 (2018). 

Variational Quantum Algorithm (VQA)

- Schematic diagram of a variational quantum algorithm:



The figure is adopted from [10]¹³.

¹³[10] Cerezo, M., Arrasmith, A., Babbush, R., Benjamin, S. C., Endo, S., Fujii, K., ..., and Coles, P. J., *Variational quantum algorithms*. Nature Reviews Physics, 3(9), (2021) 625-644.

Variational Quantum Algorithm (VQA)

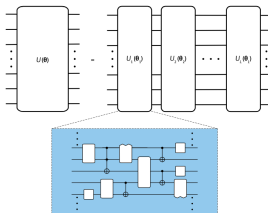
- (Ansatz) The form of the ansatz dictates what the parameters θ are, and hence how they can be trained to minimize the cost.
- For the cost function, the parameters θ can be encoded in a unitary $U(\theta)$ that is applied to the input states to the quantum circuit.

$$U(\theta) = U_L(\theta_L) \cdots U_2(\theta_2)U_1(\theta_1),$$

with

$$U_l(\theta_l) = \prod_m e^{-i\theta_m H_m} W_m.$$

Here, W_m is an unparameterized unitary and H_m is a Hermitian operator; θ_l is the l th element in θ .



Schematic diagram of an ansatz. The figure is adopted from [10].

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- [17] (Preliminary) Let H be a Hermitian matrix.

$$H = \sum_{i=1}^N \lambda_i |\psi_i\rangle \langle \psi_i|$$

where each λ_i is the eigenvalue corresponding to the eigenvector $|\psi_i\rangle$.

$$\langle H \rangle_\psi := \langle \psi | H | \psi \rangle.$$

Thus, $\langle H \rangle_\psi = \sum_{i=1}^N \lambda_i |\langle \psi_i | \psi \rangle|^2$. For (unit) state $|\psi\rangle$,

$$\lambda_{\min} \leq \langle H \rangle.$$

The above equation is known as the variational method (in some texts it is also known as the variational principle¹⁴). Note that $\langle H \rangle_{\psi_{\min}} = \lambda_{\min}$.

¹⁴Rayleigh–Ritz variational principle.

- [10] VQE is aimed at finding the ground-state energy of a Hamiltonian H . Here, the cost function is defined as $C(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$ and $|\psi(\theta)\rangle = U(\theta)|\psi_0\rangle$ for some ansatz $U(\theta)$ and initial state $|\psi_0\rangle$.
- [9]¹⁵ In practical, H can be written as a sum over Pauli operators,

$$H = \sum_{i=1}^n \sum_{\alpha \in \{x,y,z,1\}} h_{\alpha}^i \sigma_{\alpha}^i + \sum_{i,j=1}^n \sum_{\alpha \in \{x,y,z,1\}} h_{\alpha,\beta}^{ij} \sigma_{\alpha}^i \sigma_{\beta}^j + \cdots,$$

and the expectation value becomes a sum of expectations,

$$\langle H \rangle = \sum_{i,\alpha} h_{\alpha}^i \langle \sigma_{\alpha}^i \rangle + \sum_{i,j,\alpha,\beta} h_{\alpha,\beta}^{ij} \langle \sigma_{\alpha}^i \sigma_{\beta}^j \rangle + \cdots,$$

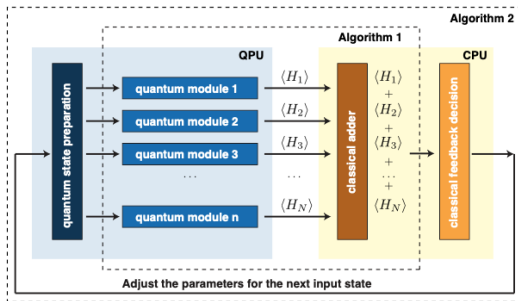
where i,j runs over the qubits that the Pauli operators acts on.

¹⁵[9] Schuld, M., and Petruccione, F., *Machine Learning with Quantum Computers*, 2021, Springer.

Variational Quantum Eigensolver (VQE)

- [1]¹⁶ QPU: quantum processing unit, efficiently calculates the expectation value of a Hamiltonian, providing an exponential speedup over conventional methods.

CPU: classical processing unit, runs an optimization algorithm, variationally computes the eigenvalues and eigenvectors of the Hamiltonian.



Architecture of the quantum-variational eigensolver. The figure is adopted from [1].

¹⁶[1] Peruzzo, A., McClean, J., Shadbolt, P., Yung, M. H., Zhou, X. Q., Love, P. J., ... and O'Brien, J. L., A variational eigenvalue solver on a photonic quantum processor, Nature communications, 5(1), (2014) 1-7.

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- A famous QVA has been proposed by Farhi and Goldstone in 2014 [5].¹⁷

The quantum approximate optimization algorithm (QAOA¹⁸), was originally introduced to obtain approximate solutions for combinatorial optimization problems.

- Consider the objective function, defined on n bit strings, is the number of satisfied clauses,

$$C(z) = \sum_{k=1}^m C_k(z)$$

where $z = z_1 z_2 \dots z_n$ is the bit string and $C_k(z) = 1$ if z satisfies clause k and 0 otherwise.

- (Example) $C_k(z) = (z_1 \wedge z_2) \vee z_3$ is fulfilled for $z = z_1 z_2 z_3 = 001$ but violated by $z = 010$.

¹⁷[5] Farhi, E., Goldstone, J., and Gutmann, S., A quantum approximate optimization algorithm. arXiv preprint (2014) arXiv:1411.4028.

¹⁸the quantum alternating operator ansatz

- [9]¹⁹ Represent this objective function by the expectation value $\langle \psi | C | \psi \rangle$ of a quantum operator, say C .

In matrix representation, the integer representation i of z defines the element $H_{i,i} = C(z)$ of the Hamiltonian and the matrix has zero off diagonal elements.

- (Example) Let $C_1(z_1, z_2) = z_1 \wedge z_2$ and $C_2(z_1, z_2) = z_1 \vee z_2$. The matrix expression of operator $C = C_1 + C_2$ would be given by

$$C = \begin{pmatrix} \langle 00 | C | 00 \rangle & \langle 00 | C | 01 \rangle & \langle 00 | C | 10 \rangle & \langle 00 | C | 11 \rangle \\ \langle 01 | C | 00 \rangle & \langle 01 | C | 01 \rangle & \langle 01 | C | 10 \rangle & \langle 01 | C | 11 \rangle \\ \langle 10 | C | 00 \rangle & \langle 10 | C | 01 \rangle & \langle 10 | C | 10 \rangle & \langle 10 | C | 11 \rangle \\ \langle 11 | C | 00 \rangle & \langle 11 | C | 01 \rangle & \langle 11 | C | 10 \rangle & \langle 11 | C | 11 \rangle \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix},$$

and contains the number of fulfilled statements on the diagonal.

- $\langle \psi | C | \psi \rangle$ is maximized if the quantum state concentrates most of its probability to basis states that represent bit strings which satisfy many clauses.

¹⁹[9] Schuld, M., and Petruccione, F., *Machine Learning with Quantum Computers*, 2021, Springer.

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- Define an unitary operator U and an operator B

$$U(C, \gamma) := e^{-\gamma C} = \prod_{k=1}^m e^{-i\gamma C_k}, \quad B := \sum_{j=1}^n \sigma_j^x. \quad (\sigma_j^x: \text{single bit Pauli operator}),$$

$$U(B, \beta) := e^{-\beta B} = \prod_{j=1}^n e^{-i\beta \sigma_j^x}$$

where β runs from 0 to π .

- The initial state $|s\rangle$ will be the uniform superposition over computational basis states:

$$|s\rangle = \frac{1}{2^{n/2}} \sum_k |k\rangle.$$

- For any integer $p \geq 1$ and $2p$ angles $\gamma_1, \dots, \gamma_p = \gamma$ and $\beta_1, \dots, \beta_p = \beta$, define the angle dependent quantum state:

$$|\gamma, \beta\rangle := U(B, \beta_p)U(C, \gamma_p) \cdots U(B, \beta_1)U(C, \gamma_1)|s\rangle.$$

- This state can be produced by a quantum circuit of depth²⁰ at most $mp + p$.
- If γ^*, β^* maximize $\langle \gamma, \beta | C | \gamma, \beta \rangle$, then sampling computational basis state from $|\gamma^*, \beta^*\rangle$ will reveal good candidates for z .
- Farhi and Goldstein have shown that

$$\lim_{p \rightarrow \infty} \max_{\gamma, \beta} \langle \gamma, \beta | C | \gamma, \beta \rangle = \max_z C(z),$$

and that even for $p = 1$, this algorithm can have useful solutions.

²⁰[10] An integer number that counts the maximum length in the circuit between the input and the output. This length is usually defined in terms of layers of gates acting in parallel.

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- [14]²¹ Consider the Ising model with the energy

$$E = - \sum_{\langle i,j \rangle} z_i z_j - \sum_i h_i z_i$$

where $\langle i,j \rangle$ means a sum over all nearest-neighbor pairs.

- (Aim) finding a low-lying state of the Ising Model.
- (Implementation $U(\gamma, C)$) Use E as C . Set

$$U(\gamma, C) = \prod_{\langle i,j \rangle} e^{-i\pi z_i z_j / 2} \prod_i e^{-i\pi \gamma h_i z_i / 2}.$$

²¹[14]



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 https://en.wikipedia.org/wiki/Quadratic_unconstrained_binary_optimization

 [https://en.wikipedia.org/wiki/Variational_method_\(quantum_mechanics\)](https://en.wikipedia.org/wiki/Variational_method_(quantum_mechanics))

Thank you for listening.